



No. of Additional Sheets

MODEL ENGINEERING COLLEGE, THRIKKAKARA, KOCHI - 682 021**"Evolve into an academy of excellence to serve the emerging knowledge society."**

To be filled by the student									
Series Test No.	1	Name of the Student		Harigovind Pangil Baija					
Date/Time	16/2/26	Class	CS	B	Sem	6	Roll No.	28	
Name of Subject with Code		HUT 300		IEFT.					
Marks to be filled by the examiner									
Questions	I	II	III	IV	V	VI	VII	VIII	IX
Marks	1.5	2	2.5			8.5	1		
Max. Marks/COs									
Maximum Marks						Marks Scored		14.05	

(Start writing from here)

Past A

1. Production Possibility Curve. Can be defined as a ~~con~~ curve of different combinations of quantities of two different commodities ~~sum~~ when technology is provided or remains ~~the~~ constant.

Assumptions:

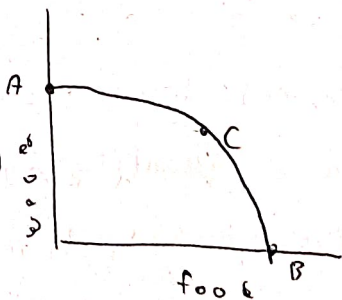
- Assume that only two commodities exist
- Technology is provided or constant
- Maximum employment is deployed.

Let water and food be the two commodities.

When only water is being produced, food is zero (A)

Similarly when only food is produced water is zero (B)

Let C be the point at which both water and food are produced equally.



Joining ABC we obtain the ~~curve~~ PPC AC

2. As per Law of Diminishing Marginal Utility, when a consumer keeps on consuming a utility, the MU will keep on decreasing.

2.

	1	2	3	4	5	6	7
TU	10	18	24	28	30	30	28
MU	10	8	6	4	2	0	-2

~~TU keeps on~~

Both TU and MU increases at first, then in the second phase it increases at a decreasing rate, at a certain point MU becomes 0, after that in the third phase MU starts decreasing.

3. Break-Even Point can be described as the point at which the firm neither makes profit nor is at a loss.

Shut down point is the point at which the firm needs to be shut down as it is not making enough revenue to sustain itself. It takes place when $\text{Revenue} = \text{Total Variable Cost}$.

4.

5. opportunity cost is the cost incurred by the firm for external needs other than production needs. for eg: Labour cost.

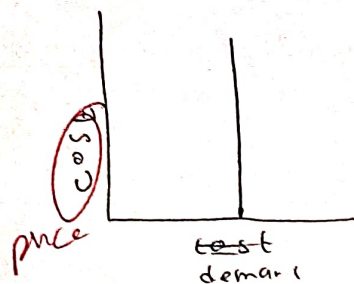
Part B

6(a) Law of demand states that "other things remaining the same, price and quantity demanded are inversely proportional." i.e. as price increases, demand drops. Some determinants of demand include:

- Price of the product
- Income of the consumer
- Price of related product.
- Climatic conditions.
- Taste and preferences of the consumer.

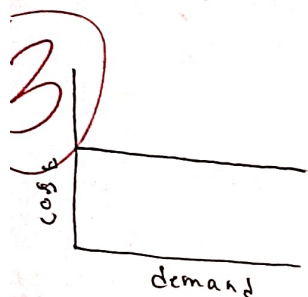
2

Perfectly elastic: - demand remains the same despite the cost.



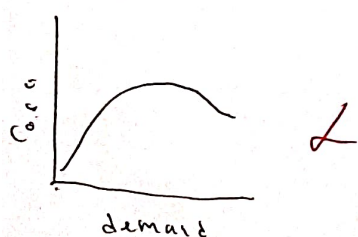
$$e = \infty$$

Perfectly inelastic: Demand changes but the cost remains the same.



$$e = 0$$

Unitary elastic: Demand changes per unit change in cost.



Relatively less elastic: Demand decreases as cost increases.



6(b) $Q_1 = 5000, Q_2 = 6000 \quad P_1 = 60, P_2 = 70$

$\Delta Q = 1000, \Delta P = -10$

Elasticity of demand = $\frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$

$= \frac{1000}{-10} \times \frac{60}{5000}$

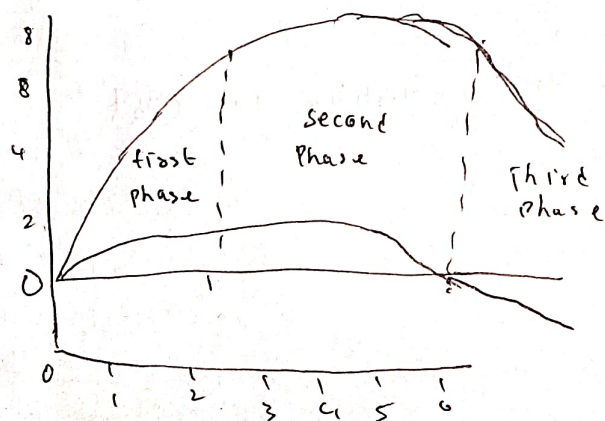
$= -\frac{6}{5} = -1.2$

$$\begin{array}{r} 1.2 \\ 5 \overline{) 6} \\ \underline{5} \\ 10 \end{array}$$

3

7(a) In ^{short} term both fixed and variable ~~cost~~ conditions are to be considered while in the long term all conditions are variable so only variable conditions are considered.

In the first phase TC increases, in the second phase TC increases at a decreasing rate and in the third phase TC starts decreasing after marginal product hits 0.



7(b)

9.